

Spot rate & forward rate practice HW

1. You are given the following term structure of interest rates.

Years	Annual Effective Spot Rate
1	3.0%
2	3.5%
3	3.9%
4	4.4%
5	5.2%

In one year, a 3-year annuity-immediate will be issued. The annuity makes annual payments of 10,000.

Calculate the present value of this annuity one year from now.

- A. 25,847
- B. 26,669
- C. 27,469
- D. 27,960
- E. 28,798

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2. An investment has the following payment structure:

1,000 payable in one year, 1,000 payable in two years, and 1,000 payable in three years.

The term structure of interest rates is:

Length of Investment	Interest Rate
1 year	6%
2 year	7%
3 year	8%

If the investment is purchased now at its value based on the term structure of interest rates, what is the overall yield rate on this investment?

- A. 7.1%
- B. 7.2%
- C. 7.3%
- D. 7.4%
- E. 7.5%

3. The one-year forward rates, deferred t years, are estimated to be:

Year (t)	0	1	2	3	4
Forward Rate	4%	6%	8%	10%	12%

Calculate the spot rate for a zero-coupon bond maturing three years from now.

- A. 4%
- B. 5%
- C. 6%
- D. 7%
- E. 8%

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4. A bond matures for its par value of 1,000 in 3 years. It pays annual coupons of 5%.

The one-, two-, and three-year annual effective spot rates are X , $X + 0.01$, and $X + 0.02$, respectively.

The annual effective yield of the bond is 6.93%.

Calculate X .

- A. 0.045
- B. 0.050
- C. 0.055
- D. 0.060
- E. 0.065

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5. The following term structure of interest rates is based on available bond market data as of 01/01/2023:

Term to maturity (years)	Spot Rate
1	3.0%
2	3.5%
3	4.0%

A three-year 1000 face amount bond with an annual coupon rate of 10% paid annually is issued on 01/01/2023.

Calculate the yield to maturity for this bond.

- A. 3.82%
- B. 3.86%
- C. 3.90%
- D. 3.94%
- E. 3.98%